AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM351
Data Management and Analysis

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 8 / April /2025

umber of Exam Pages: 5 ncluding this cover sheet)	Time Allowed: 2 Hours

nstructions:

- Total Marks: 50
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are NOT allowed.

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Part I: [5 marks]

Please select the best and most appropriate answer and write it on the answer booklet

1. The four stages of the data pipeline are: Acquisition, Preparation, Analysis and _____.

a. Archiving b. Data management c. Presentation d. Repurposing

2. The _____ character encoding encodes all 1,112,064 valid code points in Unicode using a variable number of bytes (from 1-4) and is backward compatible with ASCII code.

a. Unicode b. ASCII c. UTF-8 d. Hoffman code

3. Which are the key issues in Data Management?

a. Legal b. Control c. Curation d. All of the above

4. In the following data set: {10, 109, 10, 108, 107,15, 251, 13} the mode is: _____.

a. 107 b. 10 c. 108 d. 251

5. Spreadsheets are easily shared, possibly outside the business for which they were created which creates a problem of data _____.

a. cleansing b. sharing c. control d. acquisition

6. JSON File stands for _____.

a. JavaSript Operator Notation b. JavaScript Object Notation c. JavaServer Object Notation d. None of the above

7. In the relational model, we cannot assign NULL to an attribute that is _____.

a. part of the primary key b. a foreign key c. composite d. a non-key attribute

8. Query languages, such as SQL, which specify what data to be retrieved but not how it is to be re-trieved are referred to as: _____ languages.

a. procedural (prescriptive) b. non-procedural (descriptive c. host d. None of the above

9. A relation is in First Normal Form (1NF) if each attribute contains only _____ values, that is, it has no repeating groups of values.

a. Atomic b. Numerical c. String d. Unique

10. _____ is a referential action that specifies that in the case a referenced tuple is deleted, all referenced tuples are deleted as well.

a. Cascade b. Restrict c. Null d. Default

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Part Two: [45 marks]

Question One: [10 marks]

What will be the output of the following snippets of python code? Write down the exact output, including all brackets, parentheses, braces, quotes and commas ([], 0, {}, '', "", ,) .

1.	def factorial(n):
	  if n == 0:
	  return 1
	  else:
	  return n * factorial(n - 1)
	def main():
	  num = 5
	  factorial_result = factorial(num)
	  print("The factorial of", num, "is:", factorial_result)
	
	if __name__ == "__main__":
	  main()

2.	def is_prime(n):
	  if n <= 1:
	  return False
	  for i in range(2, int(n**0.5) + 1):
	  if n % i == 0:
	  return False
	  return True
	def main():
	  num = 17
	  if is_prime(num):
	  print(num, "is a prime number.")
	  else:
	  print(num, "is not a prime number.")
	
	if __name__ == "__main__":
	  main()

3.	[5 ,4 ,3 ,2 ,1] = numbers total = 0 :for number in numbers total += number print("The sum of the numbers is:", total)
4.	numbers = [10, 20, 30, 40, 50] total = 0 count = 0 for number in numbers: total += number count += 1 average = total / count print("The average of the numbers is:", average)

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5.	class P:
	  def __init__(self, name, age,gpa):
	  self.name = name
	  self.age = age
	  self.gpa=gpa
	p1 = P("John", 36,2.6)
	p1.age= p1.age+4
	print(p1.name +"ala")
	print(p1.gpa-12)
	print(p1.gpa)

Question Two: [15 Marks]

A library needs a database to implement a book-borrow system.

a) Normalize the given data into 1NF, 2NF and 3NF. Show the complete relations with the given data in each normal form. [10 marks]

Member_ID	Member_Name	Member_Address	Book_ISBN	Book_Name	Book_Category	Author_ID	Author_Name	Publisher_ID	Publisher_Name	Loan_Date	Return_Date
M01	Weam	Muscat	1266	C++	IT	A101	Sam	WL	Wiley	15-02-2025	22-02-2025
M12	Qais	Muscat	1098	Algebra	Math	A204	Majid	SP	Springer	23-02-2025	02-03-2025
M01	Weam	Muscat	25	Othello	English	A89	Victor	WL	Wiley	15-02-2025	22-02-2025

b) List primary keys & foreign keys (if any) of all tables in 3NF in the following format. [5 marks]

Table Name	Primary key	Foreign key

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Question Three: [10 Marks]

Consider the following relations: -

1. ARTIST (artist_id, artist_name, genre, country) - artist_id is the primary key.
2. ALBUM (album_id, album_title, release_year, artist_id) - album_id is the primary key. artist_id is a foreign key referencing ARTIST.
3. SONG (song_id, song_title, duration, album_id) - song_id is the primary key. album_id is a foreign key referencing ALBUM.

Use the above schema and write SQL queries for the following:

1. List the title and duration of all songs in the album with title 'Humour'. (3 marks)
2. List the title of all songs by the artist 'Mohammed Rafi'. (3.5 marks)
3. Update the release year of the album with album_id 25 to 2001 and its title to 'The Army'. (1.5 marks)
4. List the name of each artist and the number of albums they have released. (2 marks)

Part Three [10 marks]

Q1. What is meant by metadata and give an example where metadata can be found. (5 marks)

Q2. A student is trying to correlate whether the grades of students depend on gender (Male or Female). Identify the Dependent and Independent variables and explain your choice. (5 Marks)

End of Questions

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